

BHAVYA PATEL

Artificial Intelligence Engineer

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Professional Summary

AI/ML Engineer with 2+ years of hands-on experience building and deploying machine learning systems — spanning deep learning pipelines, LLM-powered applications, RAG workflows, and computer vision models. Completed Master of Applied Computer Science at Concordia University (2026) with practical experience in PyTorch, TensorFlow, LangChain, Hugging Face, SQL, and cloud platforms including AWS and Azure. Published researcher in computer vision (Springer, ICDSA 2024) with demonstrated results: 93.7% classification accuracy on a Vision Transformer model and 40% automation efficiency gain on an LLM pipeline. Eager to contribute to production-grade AI Engineering, Machine Learning, or Data Science teams.

Education

Concordia University <i>Master of Applied Computer Science (MAppCompSc)</i>	Sept 2024 – May 2026 <i>Montreal, QC, Canada</i>
Charotar University of Science and Technology (CHARUSAT) <i>Bachelor of Technology in Computer Engineering (B.Tech)</i>	Jun 2020 – May 2024 <i>Gujarat, India</i>

Skills

- **Languages & Frameworks:** Python, C, C++, Java, SQL, Shell Scripting, Django, FastAPI, Flutter, REST API Integration
- **AI & Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Keras, Hugging Face, LangChain, LangGraph, Natural Language Processing (NLP), Computer Vision, Deep Learning, Generative AI, Retrieval-Augmented Generation (RAG), Model Fine-tuning, Prompt Engineering, LLM Frameworks, MLOps
- **Data Science & Analytics:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Streamlit, Tableau, Power BI, Data Analytics, Data Visualization
- **Vector Databases & Storage:** Pinecone, ChromaDB, FAISS, MySQL, Vector Databases
- **MLOps & Deployment:** MLflow, Weights & Biases, Docker, GitHub Actions, CI/CD, Model Deployment, FastAPI
- **Cloud & Tools:** AWS (S3, EC2, Lambda, Bedrock, SageMaker), Azure ML, Git, GitHub, Jira, Linux, PyTest

Work Experience

Teaching Assistant – Distributed Systems <i>Concordia University</i>	Sep 2025 – May 2026 <i>Montreal, Canada</i>
• Directed and facilitated 10+ lab sessions and weekly tutorials on Distributed Systems concepts including Java RMI, CORBA, and SOAP-based Web Services for 20+ students. • Authored and conducted 10+ tutorials, 5+ laboratory sessions, 10+ page technical presentations, study guides, and programming exercises covering Java RMI, CORBA, and SOAP-based Web Services, enabling students to apply core Distributed Systems concepts through hands-on implementation. • Evaluated 50+ assignments and final projects per term, maintaining consistent grading standards and providing structured feedback within 5 days to support continuous student improvement.	

Data Science Intern <i>Blue Data Consulting & IT Services Pvt. Ltd.</i>	Dec 2023 – Jun 2024 <i>Surat, India</i>
• Architected an end-to-end AI pipeline converting 50+ PowerPoint presentations into video courses using LLMs, RAG, and Streamlit, increasing automation efficiency by 40%. • Benchmarked 4 LLMs across accuracy and generation speed metrics, achieving 25% higher script accuracy and 35% faster generation with GPT-3.5-Turbo. • Collaborated with a 5-member team to produce 10+ AI-based course modules covering LangChain, Flowise AI, and Prompt Engineering, boosting content scalability by 3x through deployment on Streamlit Cloud.	

- Designed and executed a deep learning research pipeline—extracting, preprocessing, and curating 13K+ video frames from the UCF-101 Sports Action Dataset using NumPy and Pandas for multi-sport activity recognition.
- Fine-tuned a Vision Transformer (ViT) model using TensorFlow and TensorFlow Hub, boosting classification accuracy from 84% to 94% through systematic hyperparameter tuning and advanced data augmentation.

- Prototyped 3+ mobile applications in Adobe XD, converting wireframes into responsive Flutter interfaces that accelerated iteration cycles by 50%.
- Delivered a client project—“Palm Box Cricket Booking System”—enabling real-time slot reservations and increasing user engagement by 40% through an intuitive Flutter UI/UX.

Major Projects

- Constructed and deployed an end-to-end image classification pipeline on Streamlit, training a PyTorch CNN on the MIT Places dataset (10,000 images) with advanced data augmentation and hyperparameter optimization.
- Compared baseline classifiers (Decision Tree, Random Forest, XGBoost) against the fine-tuned CNN, logging all
- Achieved 91.5% mAP on the test set, surpassing the baseline models by 16.22% in classification performance—codebase publicly available at github.com/BhavyaPatel25/Museum-View-Detection.

- Engineered an open-vocabulary segmentation benchmarking pipeline on the LVIS dataset (1,200+ categories) as Master’s capstone, creating a balanced 16,000-image subset to evaluate zero-shot object segmentation at scale.
- Generated and evaluated 140,000+ detections using a hybrid Grounding DINO + SAM2 architecture on an RTX 4090 GPU, automating mask generation, RLE encoding, and official LVIS metric computation via custom Python scripts.
- Achieved 18.6% mAP on rare categories without any LVIS training, outperforming a fine-tuned Mask R-CNN baseline (0.0% rare-category mAP)—codebase publicly available at github.com/BhavyaPatel25/Grounded-SAM2-LVIS-Benchmark.

Publications

- Co-authored and spearheaded research on a novel ViT-based deep learning approach for multi-class sports action recognition, overseeing model design, training pipeline (TensorFlow, TensorFlow Hub), and experimentation on 10,000+ video frames from UCF-101, achieving significant accuracy improvements over CNN baselines.
- Presented at ICDSA 2024, India, and published in Springer Book Series. DOI: [10.1007/978-981-96-2179-8_35](https://doi.org/10.1007/978-981-96-2179-8_35)

Activities & Leadership

- Led and coordinated 10+ social, technical, and cultural events impacting 500+ students, managing cross-functional team operations and improving coordination efficiency by 25%.

Certifications

- Fundamentals of Digital Image and Video Processing** – Coursera, Northwestern University: Core computer vision concepts including signal processing and motion estimation using Python, supporting deep learning research.
- Architecting Smart IoT Devices** – Coursera, EIT Digital: IoT system design including edge computing and secure communication protocols, relevant to real-time data pipelines and embedded AI systems.